

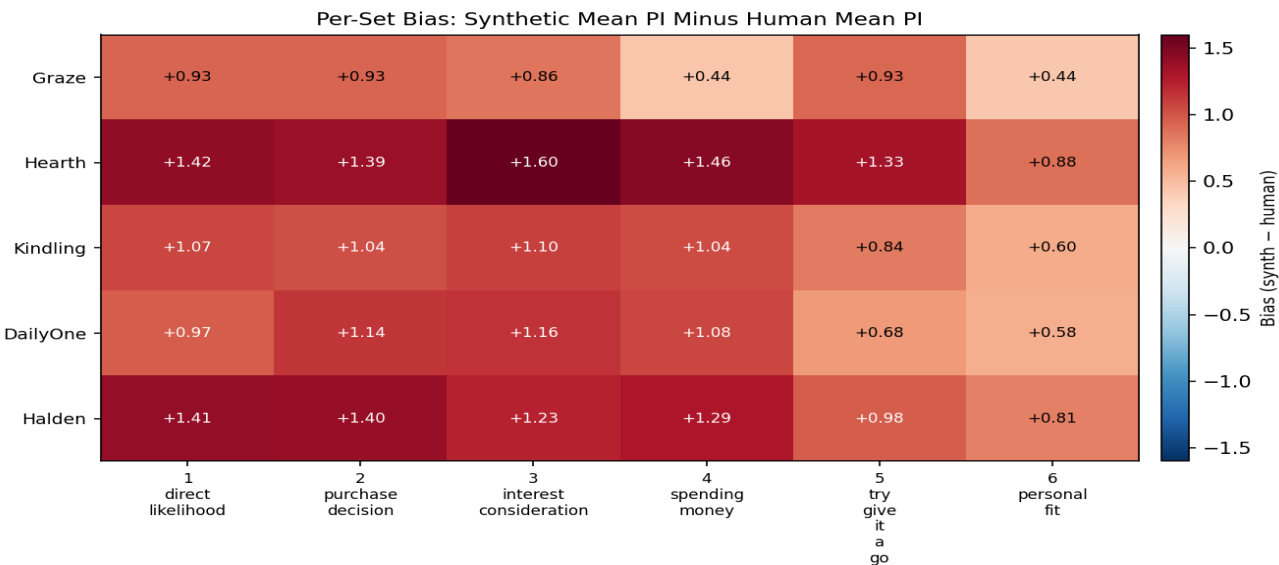
Per-Set Bias Analysis

Decomposing synthetic positivity bias by SSR reference set.

Generated: April 29, 2026

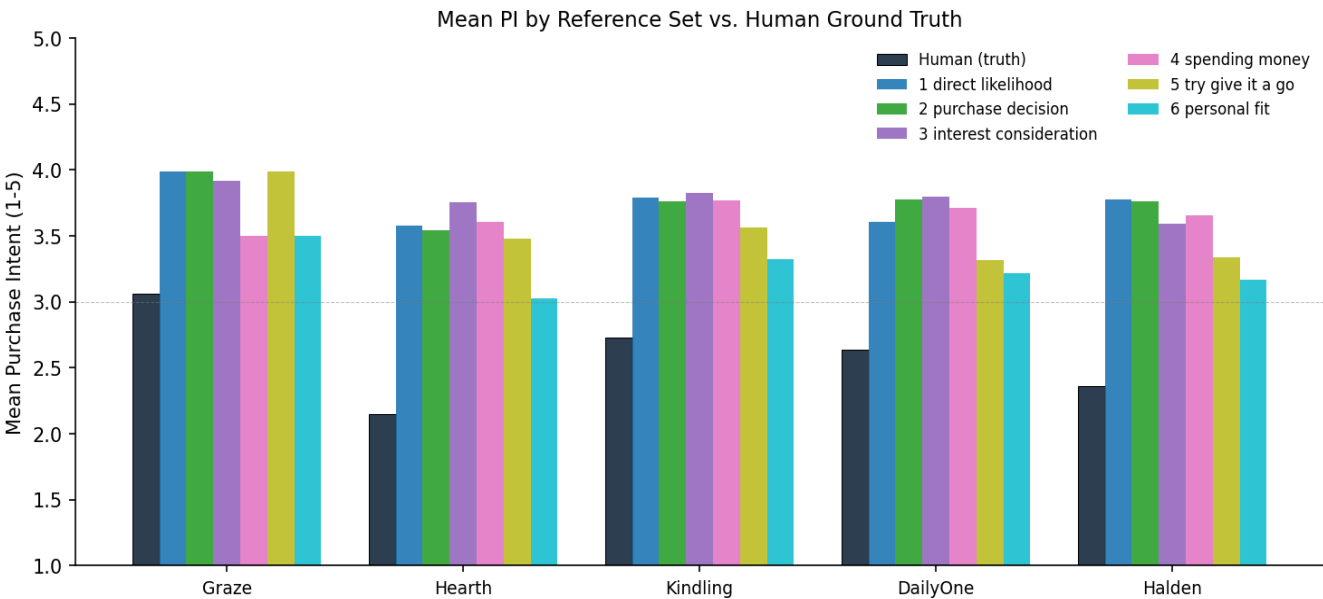
What this report does. The pipeline averages PMFs across 6 reference sets to produce its final Likert distribution. This analysis breaks that average apart and shows what each set *individually* would have produced. Sets that overshoot human ground truth more than others are candidates to drop or down-weight.

Bias matrix

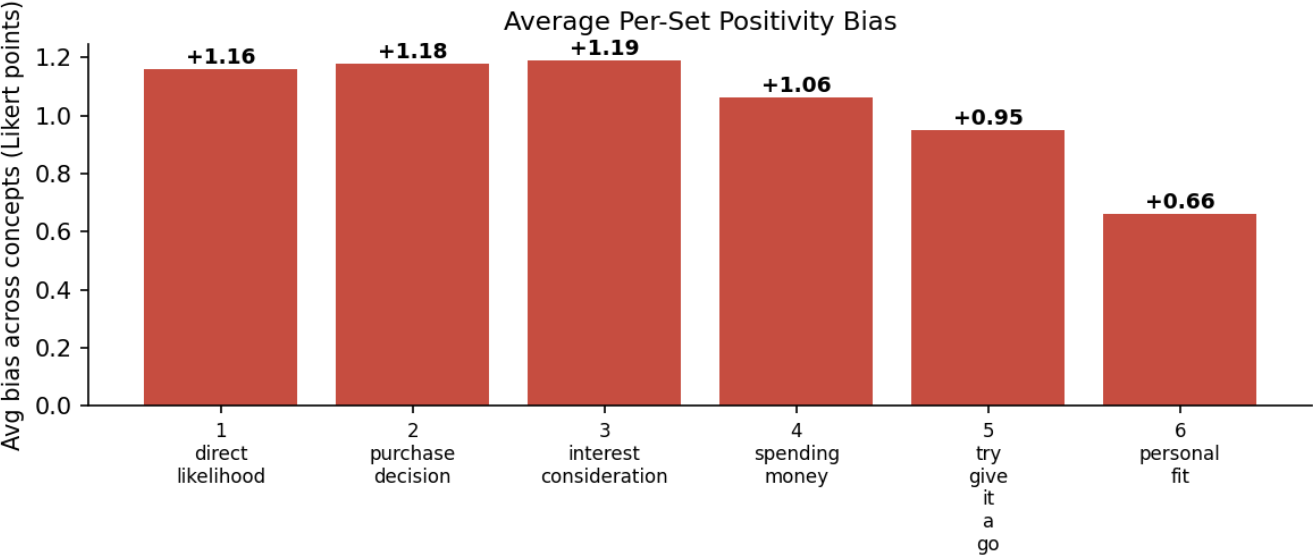


Cells show **synthetic_set_mean - human_mean** for each (concept, set) pair. Red = synthetic overshoots human; blue = synthetic undershoots. Larger absolute values = stronger bias.

Per-set means vs. human ground truth



Average bias by reference set



Reference Set	Avg Bias (Likert pts)	Verdict
3_interest_consideration	+1.19	Worst overshooter
2_purchase_decision	+1.18	
1_direct_likelihood	+1.16	
4_spending_money	+1.06	Mildest
5_try_give_it_a_go	+0.95	
6_personal_fit	+0.66	

Drop-one simulation

What happens to mean bias and ranking if we drop a single reference set and average the remaining 5?

Dropped Set	New Avg Bias	Δ vs Baseline	Ranking Preserved?
(none — baseline)	+1.03	—	✓
3_interest_consideration	+1.00	-0.03	✗ (Graze > Kindling > Halden > DailyOne > Hearth)
2_purchase_decision	+1.00	-0.03	✓
1_direct_likelihood	+1.01	-0.03	✓
4_spending_money	+1.03	-0.01	✓
5_try_give_it_a_go	+1.05	+0.02	✓
6_personal_fit	+1.11	+0.07	✓

Per-concept detail

Graze

Reference Set	Mean PI	Bias vs Human
Human (truth)	3.06	—
1_direct_likelihood	3.99	+0.93
2_purchase_decision	3.99	+0.93
3_interest_consideration	3.92	+0.86
4_spending_money	3.50	+0.44
5_try_give_it_a_go	3.99	+0.93
6_personal_fit	3.50	+0.44
Avg of all 6 sets	3.81	+0.75

Hearth

Reference Set	Mean PI	Bias vs Human
Human (truth)	2.15	—
1_direct_likelihood	3.58	+1.42
2_purchase_decision	3.54	+1.39
3_interest_consideration	3.75	+1.60
4_spending_money	3.61	+1.46
5_try_give_it_a_go	3.48	+1.33
6_personal_fit	3.03	+0.88
Avg of all 6 sets	3.50	+1.35

Kindling

Reference Set	Mean PI	Bias vs Human
Human (truth)	2.73	—
1_direct_likelihood	3.79	+1.07
2_purchase_decision	3.76	+1.04
3_interest_consideration	3.83	+1.10
4_spending_money	3.77	+1.04
5_try_give_it_a_go	3.56	+0.84
6_personal_fit	3.33	+0.60
Avg of all 6 sets	3.67	+0.95

DailyOne

Reference Set	Mean PI	Bias vs Human
Human (truth)	2.64	—
1_direct_likelihood	3.61	+0.97
2_purchase_decision	3.78	+1.14
3_interest_consideration	3.80	+1.16
4_spending_money	3.71	+1.08
5_try_give_it_a_go	3.32	+0.68
6_personal_fit	3.22	+0.58
Avg of all 6 sets	3.57	+0.93

Halden

Reference Set	Mean PI	Bias vs Human
Human (truth)	2.36	—
1_direct_likelihood	3.77	+1.41
2_purchase_decision	3.76	+1.40
3_interest_consideration	3.59	+1.23
4_spending_money	3.66	+1.29
5_try_give_it_a_go	3.34	+0.98
6_personal_fit	3.17	+0.81
Avg of all 6 sets	3.55	+1.19

Recommendations

Large spread across sets (0.53 Likert pts). **3_interest_consideration** is a clear outlier (+1.19 bias). Try dropping it and re-running the average. This may meaningfully reduce positivity bias without requiring a full pipeline re-run.

What this analysis can't tell you. Per-set bias is computed against this single 5-concept pilot. Set-level fixes that work here may not generalize to other categories or demographics. Don't permanently drop sets based on one validation run — instead, treat this as a hypothesis to test against your next validation study.